

Supplementary Information

Reducing transition costs towards carbon neutrality of China's coal power plants

Rui Wang¹, Wenjia Cai^{1,*}, Ryna Yiyun Cui², Lin Huang³, Weidong Ma⁴, Binbin Qi⁵, Jia Zhang³, Jiang Bian⁴, Haoran Li^{6,7}, Shihui Zhang¹, Jianxiang Shen¹, Xian Zhang⁸, Jiutian Zhang⁹, Wei Li¹, Le Yu^{1,10,11}, Ning Zhang¹², Can Wang⁷

Affiliations:

¹ Department of Earth System Science, Ministry of Education Key Laboratory for Earth System Modeling, Institute for Global Change Studies, Tsinghua University, Beijing 100084, China

² Center for Global Sustainability, School of Public Policy, University of Maryland, College Park, MD, USA

³ Microsoft Research AI4Science, Beijing 100080, China

⁴ Microsoft Research Asia, Beijing 100080, China

⁵ State Key Laboratory of Petroleum Resources and Prospecting, China University of Petroleum (Beijing), Beijing 102249, China

⁶ China Electric Power Planning & Engineering Institute, Beijing 100120, China

⁷ State Key Joint Laboratory of Environment Simulation and Pollution Control (SKLESPC), School of Environment, Tsinghua University, Beijing 100084, China

⁸ The Administrative Centre for China's Agenda 21, Ministry of Science and Technology, Beijing, China

⁹ Green Development Institute, Beijing Normal University, Beijing 100875, China

¹⁰ Ministry of Education Ecological Field Station for East Asian Migratory Birds, Beijing 100084, China

¹¹ Tsinghua University (Department of Earth System Science)- Xi'an Institute of Surveying and Mapping Joint Research Center for Next-Generation Smart Mapping, Beijing 100084, China

¹² Department of Electrical Engineering, Tsinghua University, Beijing 100084, China.

*Corresponding author. Email: wcai@tsinghua.edu.cn

Contents

Supplementary Note.....	3
Supplementary Note 1: Coal power plant dataset	3
Supplementary Note 2: Biomass resource dataset	6
Supplementary Note 3: Carbon storage dataset	7
Supplementary Method	8
Supplementary Method 1: Carbon emission reduction pathways (CERPs).....	8
Supplementary Method 2: Mathematical formulation of the plant-level dynamic optimization model.....	10
Supplementary Method 3: Transition costs of the coal power sector.....	14
Supplementary Figures	15
Supplementary Tables	19
Supplementary References.....	24

List of Supplementary Figures

Supplementary Figure 1 Comparison of provincial installed capacity between the government statistics and the plant-level dataset..	5
Supplementary Figure 2 The spatial distribution of biomass resources sourced from agricultural and forestry residues.	6
Supplementary Figure 3 The carbon storage capacity of China's deep saline basin.....	7
Supplementary Figure 4 The carbon emission reduction pathways (CERPs) under different cumulative carbon emission reduction targets (CuERs).	9
Supplementary Figure 5 Comparison between the carbon emission pathways of coal power in IPCC AR6 database and the carbon emission reduction pathways in this study.....	9
Supplementary Figure 6 Coal power generation constraints for each province.....	15
Supplementary Figure 7 Uncertainty analysis for the techno-economic parameters..	16
Supplementary Figure 8 Spatial distribution of coal power plants in China.	17
Supplementary Figure 9 The characteristics of coal power plants in each province.....	18

List of Supplementary Tables

Supplementary Table 1 The benchmark for clean and efficient utilization of coal	3
Supplementary Table 2 The statistics of coal plants for each province.....	19
Supplementary Table 3 Techno-economic parameters for the plant-level optimization model.	20
Supplementary Table 4 Statistics of the uncertainty analysis for the techno-economic parameters.	21
Supplementary Table 5 Carbon emission parameters of the coal power plants and retrofit technologies.	22
Supplementary Table 6 Coal power generation constraints for each province.	23

Supplementary Notes

Supplementary Note 1: Coal power plant dataset

(1) already existing coal power plants.

Currently, there are many publicly available coal power plant databases, such as the Global Coal Plants Tracker, S&P Global Platt's World Electric Power Plants (WEPP) database, and WRI's global database of power plants. Based on these publicly available databases, we also integrated many structured and unstructured data sources to further update and revise the coal power plant data in China. This dataset includes information on installed capacity, operating hours, unit type, commissioning year, spatial location, and coal consumption per unit of electricity. Further details and its application can be found in the research articles published by Li et al^{1,2}.

(2) newly constructed and planned coal power plants.

For the newly constructed and planned coal power plants after 2020, we use the data from Global Coal Plants Trackers (GEM) directly, as it is timely (updated every month), free, and rich in details. The GEM database includes lots of details about coal plants, such as their current status, start year, combustion technology, latitude, longitude, capacity factor, etc. However, some missing values remain, such as the commissioning years, unit coal consumption, operation hours, etc. Here, we outline our approach to address these issues.

a. The missing value of the commissioning year.

In GEM, the status of the newly built plants includes 'announced,' 'construction,' 'permitted,' 'pre-permitted,' and 'operating'. By consulting the expertise of the Electric Power Planning Research Institute, the coal power projects that are more likely to be built are the plants labeled as 'operating' and 'construction', considering China is very likely to stop building new coal plants after 2030. Therefore, we only considered these types of coal plants and assigned 2024 and 2027 as their commissioning years for these two types of coal plants with null values.

b. The missing value of unit coal consumption.

we fill in the missing values based on its combustion technology characteristics by following the guidelines outlined in the "Benchmark Levels and Benchmark Levels in Key Areas of Clean and Efficient Utilization of Coal (2022 Edition)" issued by the National Development and Reform Commission of China. We assume that all existing units have reached the baseline level, and the specific assignments are described in Supplementary Table 1.

Supplementary Table 1 The benchmark for clean and efficient utilization of coal

Combustion technology	Capacity (MW)	Basic level	Unit
Ultra-supercritical	>=600	285	g / kWh
	<600	293	g / kWh
Supercritical	>=300	300	g / kWh
	<300	308	g / kWh
Subcritical	>=300	314	g / kWh
	<300	323	g / kWh
CFB	/	290	g / kWh
IGCC	/	270	g / kWh
Ultra-supercritical/ccs	/	270	g / kWh

Note: this standard is issued by the National Development and Reform Commission of China in 2022 (https://www.ndrc.gov.cn/xxgk/zcfb/tz/202205/t20220510_1324482.html).

c. The missing value of the operating hours.

We use the national average operating hours to replace the null value, which is 4611.83 hours, sourced from Zhou et al³.

(3) Before serving as the model's input.

a. aggregate coal power units as plants. In the study, we aggregate the units that have the same company name, spatial location, commissioning year, technology status, operating hours, and coal consumption as the same plant.

b. Calculate the potential distance from coal power plants to the CO₂ storage site. As the pipelines of carbon transportation have not been constructed yet, we may not be able to calculate the transportation distance accurately. Therefore, in this study, we make some simplifications. First, we achieve the potential carbon storage sites based on the carbon storage area (Supplementary Note 3). Then we help each coal plant find its nearest carbon storage site and calculate their Euclidean distance. As the routes of pipeline networks or truck transportation are often circuitous, we assume the tortuosity factor as 3.7⁴⁻⁶ and multiply this factor by the Euclidean distance to reflect the carbon transportation distance if coal plants choose CCS technology.

c. As we assume the lifetime of the coal plants is 40 years, we exclude the coal plants that are committed before 1985.

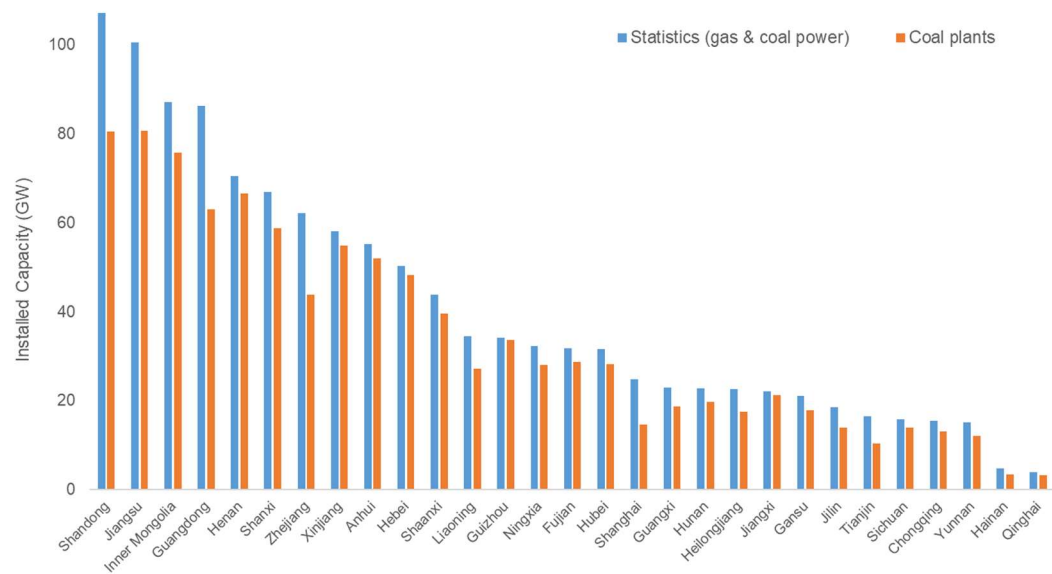
(4) Data validation.

According to the statistics released by the China Electricity Council, by the end of 2019, the installed capacity of coal power is 1040.6 GW. Their analysis in 2021 suggests that China's coal power capacity will reach 1230 GW and 1260 GW in 2025 and 2030, respectively, with a peak in 2030 (<https://news.bjx.com.cn/html/20220126/1201494.shtml>).

Our dataset aligns closely with these statistics. By the end of 2019, the capacity of our coal power dataset is 988.1 GW, covering 95% of the national installed capacity. Taking into account the 40-year design lifetime, natural retirements of the coal plants, and the emergence of new installed capacity, our database shows that the installed capacity of coal power plants will reach 1,218 GW and 1,360 GW in 2025 and 2030, respectively. Given the significant uncertainty surrounding future development, we believe this projected installed capacity is reasonable, affirming the validity of our database.

To improve the credibility of the data, we compare the official data values of each province with the statistics values from China's electricity statistics yearbook. Since China's electricity statistics yearbook does not distinguish between coal and gas power but refers to all as thermal power at the provincial level⁷, we can only compare the provincial thermal power installed capacity with our dataset by the end of 2020. Given that natural gas is expensive in China and its installed capacity only accounts for 8.3% of the thermal install capacity, these provincial statistics can still serve as a reference to validate the credibility of our coal plant dataset.

Supplementary Figure 1 shows that our dataset aligns well with provincial statistics. The discrepancies between our dataset and the statistics may arise from three possible reasons: (1) the statistics include gas power installed capacity; (2) we exclude coal plants with an installed capacity smaller than 10MW, in line with policy requirements to phase out smaller and less efficient thermal power plants⁸; and (3) inherent statistical errors in both datasets.

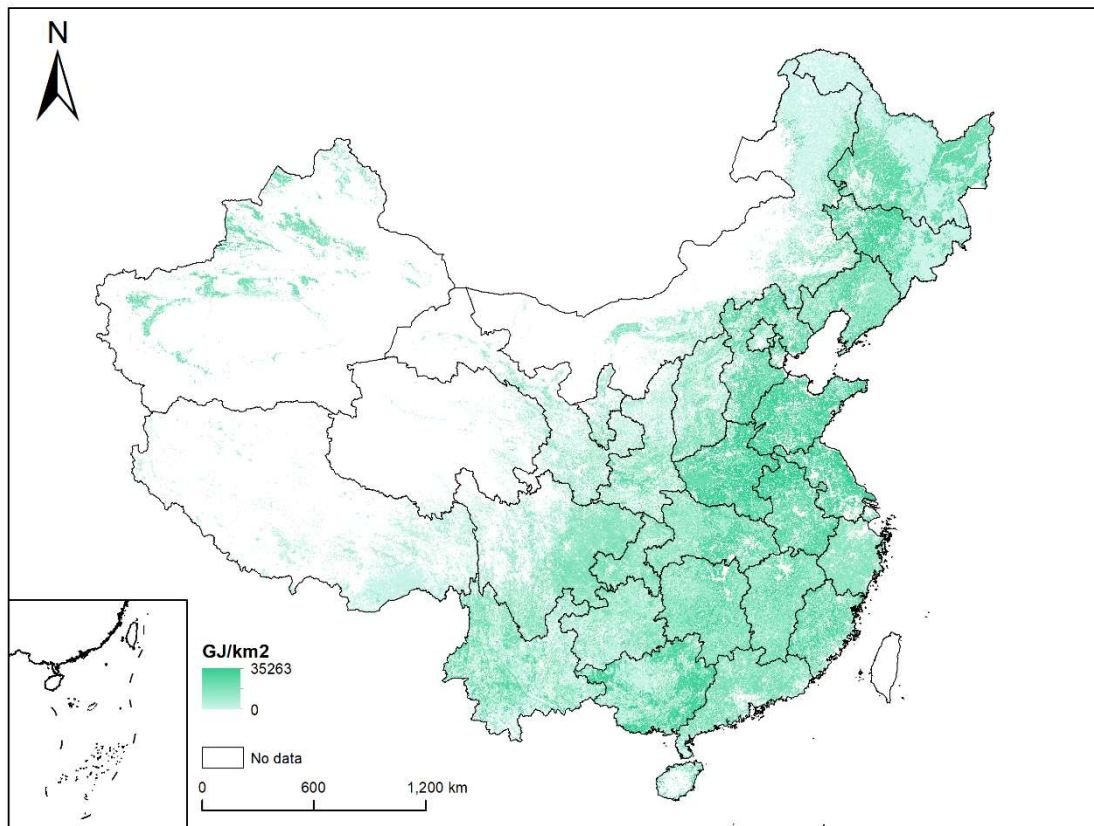


Supplementary Figure 1 Comparison of provincial installed capacity between the government statistics and the plant-level dataset.

Supplementary Note 2: Biomass resource dataset

As energy crops are not available currently, biomass resource potential sourced from agricultural residues and forestry residues are used as input for this model. The assessment of agricultural residues considers 9 types of agricultural products, including maize, canola, wheat, peanuts, other grain, soybeans, cotton, and potatoes. The assessment of the forestry residues consists of the residues from wood exploitation and process (including commercial wood, fuel wood, and logs), bamboo exploitation and process, and forest logging (including fuel forest, timber forest, protection forest, economic forest, special forest, shrub forest, and sparse forest). These data are publicly available and produced by Wang et al⁹.

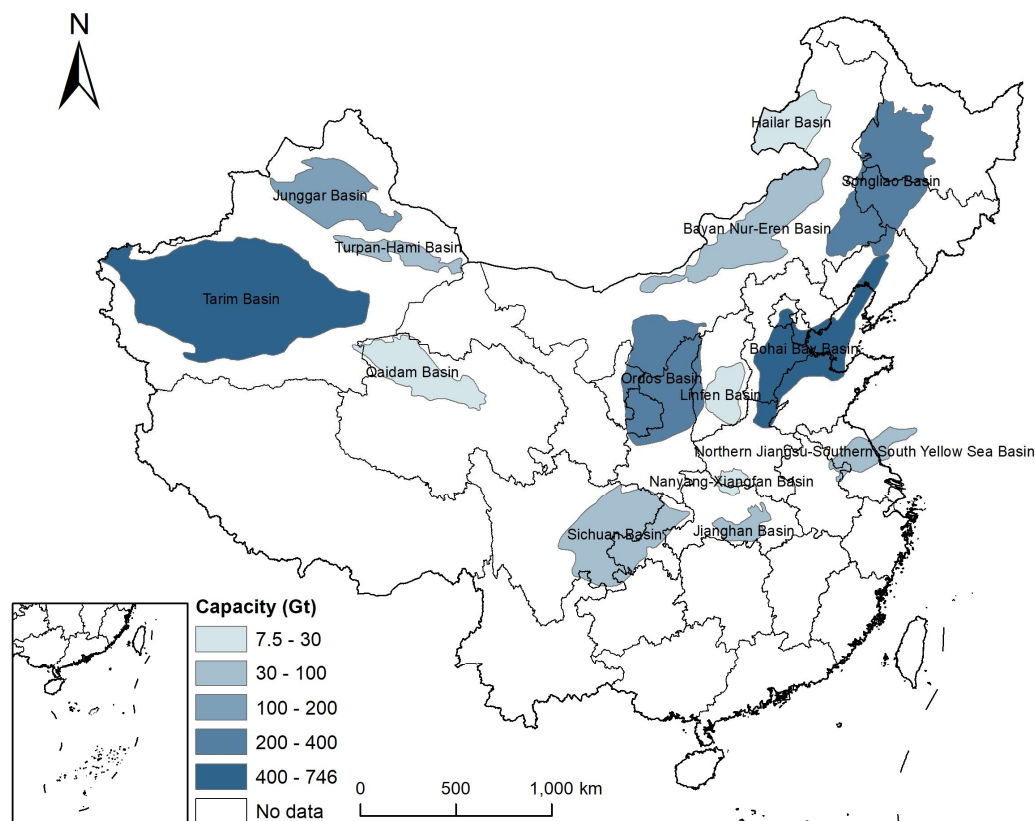
According to our assessment, the total amount of biomass resource potential is 17.9 EJ, among which Heilongjiang, Henan, and Shandong provinces have the maximum potential, with 1.42, 1.39, and 1.29 EJ, respectively. Their spatial distribution map can be found in Supplementary Figure 2.



Supplementary Figure 2 The spatial distribution of biomass resources sourced from agricultural and forestry residues.

Supplementary Note 3: Carbon storage dataset

(1) carbon storage dataset. The total carbon storage potential in China is 2,449.4 Gt, among which the oil basin can store 5.1 Gt, the gas basin can store 24.3 Gt, and the deep saline basin can store 2,420 Gt CO₂. This study uses the onshore saline basin as the carbon storage area, with a total storage potential of 2,317.1 Gt, accounting for 94.6% of the total storage potential. Deep saline basins are distributed in the Tarim Basin, Bohai Bay Basin, Songliao Basin, Ordos Basin, etc. The spatial distribution map of carbon storage potential can be found in Supplementary Figure 3.



Supplementary Figure 3 The carbon storage capacity of China's deep saline basin.

(2) potential carbon storage site. This study divides the carbon storage area with county administrative data. Then we choose the centroid of each county as the potential storage site. The storage capacities of these sites are calculated according to the ratio of the divided area to the original area. In the end, we use the k-mean method to cluster these sites according to their longitude, latitude, and storage capacity. 40 potential storage sites are used to store the captured carbon from coal power plants¹⁰.

Supplementary Methods

Supplementary Method 1: Carbon emission reduction pathways (CERPs)

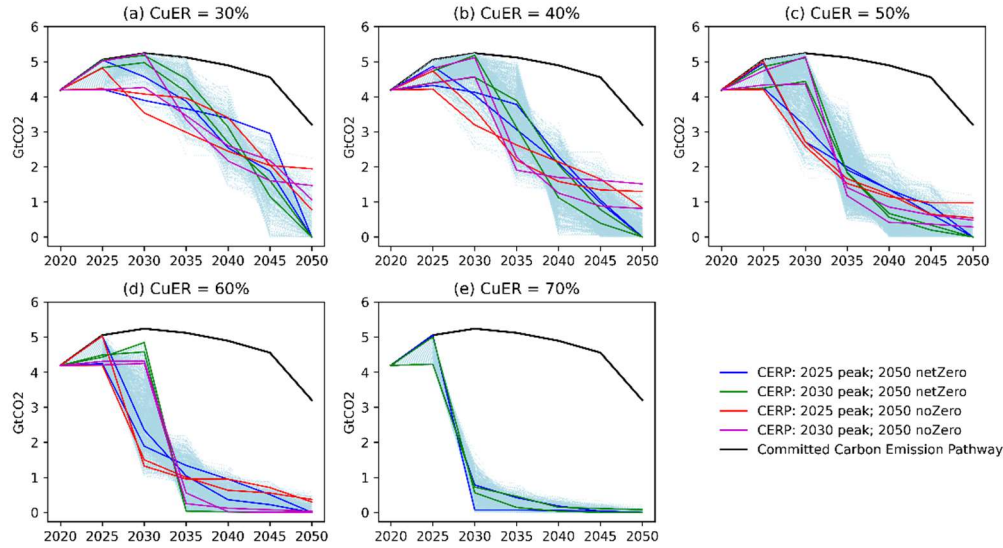
(1) CERP generation method and framework

We assume that the original lifespan for a coal power plant is 40 years. Therefore, based on the information on the commissioning year, we can calculate their carbon emission pathways as shown in Supplementary Figure 4. If these coal power plants do not take any low carbon transformation action, their cumulative carbon emission amount will reach 161.2 Gt. As there is no consensus on the cumulative carbon emission reduction target (CuER), we investigate the scenarios in the ADVANCE database^{11,12} and try to find a range for this target. We compare the carbon emission amount of the power sector's residues fossil fuel in the 'NoPolicy' scenario with the carbon emissions in the '2020_1.5C', '2030_1.5C', '2030_Price1.5C', '2030_3xPrice1.5C', '2020_Med2C', '2030_Med2C', '2020_WB2C', '2030_WB2C' scenarios. We find that the cumulative carbon emission reduction amount could range from 30% to 70%. Therefore, we use 10% as a step for the sensitivity analysis.

Even though there is no specific target for the CuER, China has set the target of reaching carbon peaking in 2030 and achieving net-zero in 2060 for the entire economy and society. As the power sector is easier to achieve carbon neutrality and must achieve low-carbon transition in advance of the manufacturing sector, we temporarily assume the coal power sector can achieve carbon peaking time in 2025 or 2030, and achieve net-zero in 2050 or postpone the net-zero time. Besides, as the carbon emission amounts in the carbon peaking time are widely discussed, we categorize the carbon peaking time into high peak and low peak. By combining these four carbon peaking scenarios (high peak in 2025, low peak in 2025, high peak in 2030, and low peak in 2030) and two net-zero times (net-zero in 2050 or postpone 2050 net-zero time), we can achieve 8 carbon emission reduction pathways (CERPs), and label them "25hp-50zero", "25lp-50zero", "30hp-50zero", "30lp-50zero", "25hp-post50zero", "25lp-post50zero", "30hp-post50zero", and "30lp-post50zero".

To achieve different curve types of CERPs under each CuER, we employ the Monte Carlo method for simulation. First, we calculate the carbon emission curve under the business-as-usual scenario based on the coal power plants dataset. Starting with the carbon emission levels in 2020, we calculate the carbon emission gap between 2020 and 2025. By randomly selecting a value within this gap, we can determine an appropriate increment for the 2025 carbon emission target relative to 2020. Similarly, if it is a CERP with a carbon peaking time in 2030, we can continue to choose a value between the carbon emission target set in 2025 and the carbon emission in 2030 under the BAU scenario, and then add this increment to the carbon emission target in 2025. If we want to control the CERP going down, the carbon emission target set in the next year can directly choose a value between the carbon emission target in the previous year and zero. By simulating this process thousands of times, we can find the line collections corresponding to our 8 types of CERPs. In the end, by calculating the cumulative carbon emission differences between these CERPs and the BAU carbon emission curves, we can filter out the curves that satisfy different CuER targets. It is important to note that the curve will fail to reach the 70% CuER target if the curves peak in 2030. Consequently, the "25lp-50zero", "25hp-50zero", "25hp-post50zero", and "25lp-post50zero" CERPs are the only viable options for 70% CuER (Supplementary Figure 4).

We acknowledge that for the same CuER, even the same types of CERPs (for example "25hp-50zero") will correspond to multiple CERPs. However, due to the computational complexity and time constraints, with our model requiring 2~4 hours to run considering numerous integer and continuous variables across 6 decision-making periods, exhaustive calculation of all CERPs is impractical. Therefore, for each type of CERP, we manually select a representative curve based on its curve shape (smoother and more characterful). For example, for the "25hp-50zero" CERPs collection, we arrange the carbon emission targets for 2025 in descending order, plot these lines, and select the curve exhibiting the smoothest trajectory.

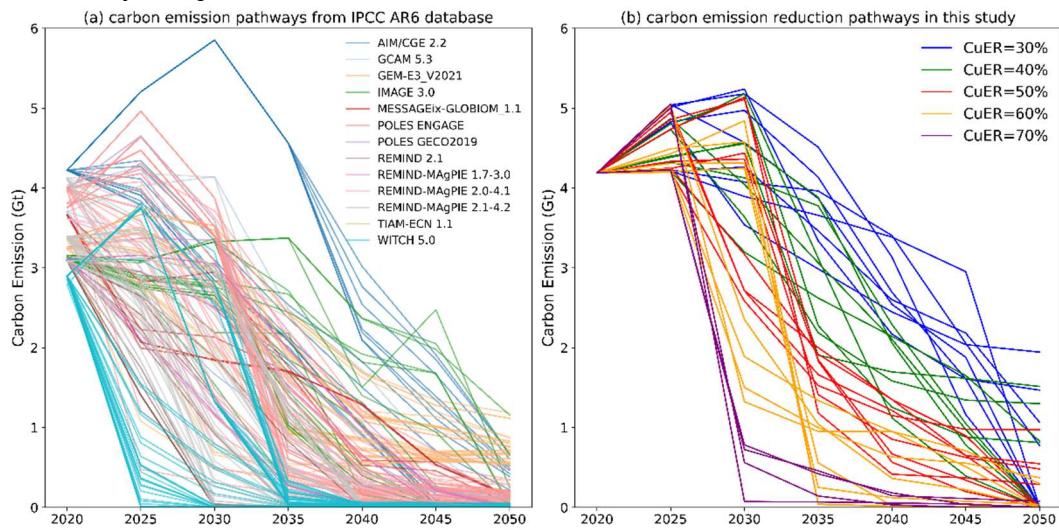


Supplementary Figure 4 The carbon emission reduction pathways (CERPs) under different cumulative carbon emission reduction targets (CuERs). Panels (a) to (e) represent the carbon emission reduction pathways under cumulative emission reduction targets from 30% to 70%.

(2) The credibility of CERPs

Our assumptions can be supported by the Integrated Assessment Models (IAMs) and the scenario database produced by Intergovernmental Panel on Climate Change in their sixth assessment report (IPCC AR6)¹³. We extracted the coal power scenarios from the IPCC AR6 scenario dataset, contributed by various Integrated Assessment Models such as AIM/CGE, GCAM, IMAGE, GEM, REMIND, POLES, TIAM, WITCH, and others. The database includes scenarios based on energy units, divided into coal electricity with and without CCS. To align with our study's assumptions, we converted these scenarios by assuming the emission factor of coal without CCS is 801.7 g CO₂/kWh and the carbon capture rate of CCS is 90%, resulting in an emission factor of 80.17 g CO₂/kWh for coal electricity with CCS.

To limit global temperature below 2°C or 1.5°C, these models indicate a gradual phase-down of coal emissions from 2020 to 2050. Our assumptions align closely with these scenarios as shown in Supplementary Figure 5. Moreover, the assumptions allow for a deeper exploration of China's carbon emission peaks, carbon neutrality timelines, and the cumulative emission reduction targets achievable by coal plants.



Supplementary Figure 5 Comparison between the carbon emission pathways of coal power in IPCC AR6 database and the carbon emission reduction pathways in this study. **a** Carbon emission pathways from the database of Intergovernmental Panel on Climate Change (IPCC) Sixth Assessment Report (AR6)¹³. **b** Carbon emission reduction pathways developed in this study.

Supplementary Method 2: Mathematical formulation of the plant-level dynamic optimization model.

A carbon-constrained plant-level dynamic optimization model is developed to explore how coal power plants can achieve low carbon transformation among multiple emission reduction technologies under a given carbon emission reduction pathway. The model inputs consist of the attribute information and spatial locations of coal power plants, a biomass resource potential map, a carbon storage potential map, and a carbon emission reduction pathway (CERP). This model takes into account four types of emission reduction technologies, including biomass and coal co-firing technology (BE), carbon capture and storage technology (CCS), flexibility operation technology (Flex), and compulsory retirement (CR). If the coal power plant has been retrofitted with BE and CCS technology, it can be regarded as a BECCS plant. The carbon removal ability of the plant depends on the amount of biomass resource it takes. Except for these four technologies, coal power plants can also remain as coal power plants (PP) or natural retirement (NR). In the end, the model can provide decisions and assessment results for each coal power plant, including technology choice, carbon emission amount, the net present value, biomass consumption, and the ultimate lifetime. This model is a recursive dynamic decision-making model, in which coal power plants make decisions every five years and the subsequent decision-making process should consider the last period status. Once the decisions are made, new investments in the equipment will happen. Thus, the low carbon transformation process for each coal power plant is irreversible but can be upgraded gradually. This model can be applied to any other region. We take China as an example to conduct the research.

Annotation

i represents biomass collection points i
 j represents coal power plant j
 k represents retrofit option k , $k=1,2,3,4$, corresponding to BE, CCS, flexible retrofit, and compulsory retirement.
 I_j install capacity of plant j
 H_j original operating hours of plant j
 P electricity price
 $Inv_{j,l}$ investment calculation function of plant j in year l
 $OM_{j,l}$ operation and maintenance cost calculation function of plant j in year l
 $Fuel_{j,l}$ fuel cost calculation function of coal power plant j in year l
 Dep_j depreciation cost calculation function of coal power plant j
 $Emiss_{j,l}$ carbon emission calculation function of coal power plant j in year l
 E_l^{CERP} the carbon emission target in year l from the carbon emission reduction pathway (CERP).

Variable

$x_{i,j,l}$ the amount of biomass resources transported from i to j
 $y_{j,k,l}$ whether coal plant j chooses the retrofit option k ; $k=1, 2, 3, 4$, corresponding to BE, CCS, Flexibility operation, and Compulsory retirement
 $f_{j,l}$ the rate of change in current operating hours of coal power plant j in year l

Model

The objective of the model is to maximize the net present values (NPV) of the coal plants from commissioning to retirement. The calculation of the NPV includes revenues and generation costs as described in supplementary equation (1). The revenues depend on the installed capacity of the coal plants, operating hours, and the electricity price. While the costs depend on the investment cost, operation and maintenance (OM) costs, fuel costs, and depreciation costs. If a coal plant j chooses the k th retrofit technology in year l , it will cause changes in investment costs, OM costs, and depreciation costs. The fuel costs will also change, which depend on its purchase amounts of coal, biomass, and actual electricity generation amount.

$$\max \sum_j \left\{ \sum_l \left(\frac{I_j * H_j * f_{j,l} * P}{(1+r)^l} \right) - \sum_l \left[\frac{Inv_{j,l}(y_{j,k,l}) + OM_{j,l}(y_{j,k,l}) + Fuel_{j,l}(x_{i,j,l}, f_{j,l}, Dis_{i,j}, Dis_{j,s}, H_j, I_j) - De_{j,l}}{(1+r)^l} \right] \right\} \quad (1)$$

Constraints

(1) Biomass resource supply constraint. If a coal power plant chooses BE or BECCS technology, it will need to collect biomass resources from nearby regions. We conduct an assessment of biomass resource potential at a 1km resolution, as detailed in Supplementary Note 2. Subsequently, we divide the whole area into 20km * 20km grid cells, set the center of each grid cell as potential biomass collection points (BCPs), and summarize the biomass resource amount fall in each grid as the BCP's supply capacity. Each BCP can provide biomass resources to multiple coal power plants, but their supply amount cannot exceed their supply capacity as described in supplementary equation (2).

$$\sum_j x_{i,j,l} \leq S_{i,l}, \forall i, l \quad (2)$$

(2) Coal power plant demand constraint. Coal power plants should not take biomass resources that exceed their demand as shown in supplementary equation (3). This demand is decided by installed capacity and the actual operating hours. As we take flexibility operation as one of the retrofit options, the changes in operating hours should also be taken into account. Therefore, the right-hand side of the constraints should include the installed capacity, original operating hours, and the change rate of the operating hours. The left-hand side of the constraints represents the total amount of biomass resource collected by coal power plant j from different BCPs in year l .

$$\sum_i x_{i,j,l} \leq I_j * H_j * f_{j,l}, \forall j, l \quad (3)$$

(3) Carbon emission constraints. The country will update its carbon emission reduction target every five years. The total carbon emission amount of the coal power plants should be smaller than the national carbon emission reduction target. In the model, the calculation of the carbon emission for one plant should also consider whether it has taken the BE or CCS retrofit in the past. If a coal plant has been retrofitted with the BE technology, even the $y_{j,be,l}$ in the current decision-making year equals zero, it still can collect biomass resources and replace them with coal. Similarly, if the coal plant has taken the CCS retrofit, it will be able to consistently capture the carbon emission from the coal plant once it adopts the retrofit action. Obviously, the integer variable of $y_{j,be,l}$, $y_{j,ccs,l}$, and $y_{j,retire,l}$ should be taken into the function as they will decide whether the coal plant will adopt the mitigation action this year as shown in supplementary equation (4). As the carbon emission is also related to operation hours, the continuous variable $f_{j,l}$ should also be taken into account.

$$\sum_j Emiss_{j,l}(y_{j,be,l}, y_{j,ccs,l}, y_{j,retire,l}, f_{j,l}, I_j, H_j, x_{i,j,l}) \leq E_l^{CERP} \quad (4)$$

(4) Electricity constraints. To ensure the security and stability of the power system, a proportion of coal power will remain to provide flexible generation resources and enable the integration of variable renewable energy. Therefore, we collect a provincial coal power generation scenario provided by a power system model (GTEP), which has considered daily power system dispatch at the micro level, the expansion of renewable energy, and the long-term technology cost changes³. This scenario will be used as the constraint for the low-carbon transition of the coal power plants. If coal plants are retrofitted with BE-related technologies, the electricity generated by biomass will also be taken into account as this kind of electricity is stable and provided by the coal power plants as shown in supplementary equation (5).

$$\sum_j I_j * H_j * f_{j,l} \geq Elec_l \quad (5)$$

(5) The constraints for flexibility operation technology.

Only if a coal power plant chooses flexibility operation, it can reduce the operation hours. Therefore, if $y_{j,k=flex,l}$ equals to zero, $f_{j,l}$ needs to be larger than 1. In converse, $f_{j,l}$ will not meet constraints as it should be larger than an infinitely larger number M as shown in supplementary equation (6).

$$f_{j,l} \geq (1 - M * \sum_{t=0}^{retrofitYr} y_{j,k=flex,t}), \forall j \quad (6)$$

Although a coal plant can reduce the operation hours after the flexibility operation, it cannot reduce operation hours without any limitations. Based on the field investigation, we assume that the lower limit of operating time is 1500 hours as shown in supplementary equation (7).

$$f_{j,l} * H_j \geq y_{j,k=flex,l=retrofitYr} * 1500, \forall j \quad (7)$$

Besides, if a coal plant has taken the flexibility operation in the past, it will always be equipped with the ability to provide flexible electricity resources. Therefore, each coal plant only needs to take less than or equal to one time of flexibility operation retrofit during its lifetime as shown in supplementary equation (8).

$$\sum_{l=0}^{retrofitYr} y_{j,k=flex,l} \leq 1, \forall j \quad (8)$$

(6) The constraints for biomass and coal co-firing technology.

If a coal power plant has not taken BE retrofit in and before year l , it cannot replace coal with biomass for emission reduction. The mathematical description of this constraint is provided in supplementary equation (9).

$$x_{i,j,l} \leq M * \sum_{l=0}^{retrofitYr} y_{j,k=be,l} \quad \forall j, l \quad (9)$$

If a coal power plant has taken compulsory retirement technology, it cannot collect biomass resources for emission reduction, as described in supplementary equation (10).

$$x_{i,j,l} \leq M * (1 - y_{j,k=retire,l}) \quad (10)$$

A coal plant can only take BE retrofit less than or equal to one time during its lifetime, as described in supplementary equation (11).

$$\sum_{l=0}^{retrofitYr} y_{j,k=be,l} \leq 1, \forall j \quad (11)$$

(7) The constraints for compulsory retirement.

If a coal plant chooses compulsory retirement, it will not be able to provide electricity and its operation hours should equal to zero, as described in supplementary equation (12).

$$f_{j,l} \leq M * (1 - y_{j,k=cr,l}), \forall j, l \quad (12)$$

A power plant can only be compulsorily retired less than or equal to one time during its lifetime, as described in supplementary equation (13).

$$\sum_{l=0}^{retrofitYr} y_{j,k=cr,l} \leq 1, \forall j \quad (13)$$

If a power plant chooses compulsory retirement, it cannot choose other retrofit technologies, such as flexibility operation, biomass and coal co-firing, or carbon capture and storage technology. The mathematical descriptions of these constraints are provided in supplementary equation (14) to (16).

$$y_{j,k=flex,l} \leq (1 - y_{j,k=retire,l}), \forall j \quad (14)$$

$$y_{j,k=be,l} \leq (1 - y_{j,k=retire,l}), \forall j \quad (15)$$

$$y_{j,k=ccs,l} \leq (1 - y_{j,k=retire,l}), \forall j \quad (16)$$

(8) The constraints for carbon capture and storage technology

A power plant can only be retrofitted with CCS technology less than or equal to one time during its lifetime, as described in supplementary equation (17).

$$\sum_{l=0}^{retrofitYr} y_{j,k=ccs,l} \leq 1, \forall j \quad (17)$$

(9) The constraints for other variables.

The amount of biomass resources transported from the biomass collection point to the coal power plant should be larger than or equal to zero, as described in supplementary equation (18). The variable indicates whether choosing this technology should be binary, as described in supplementary equation (19). $f_{j,l}$ should be a positive number, as described in supplementary equation (20). The operating hours of a coal plant should be smaller than 8000, as described in supplementary equation (21).

$$x_{i,j,l} \geq 0 \quad (18)$$

$$y_{j,k,l} = \{0,1\}, \forall j, k, l \quad (19)$$

$$0 \leq f_{j,l}, \forall j, k, l \quad (20)$$

$$f_{j,l} * H_j \leq 8000, \forall j, k, l \quad (21)$$

Calculation function.

(1) the calculation of the investment cost.

The investment cost should include the original investment cost of the coal power plant. If it chooses BE, CCS, or flexibility operation, some additional costs should be added, as shown in supplementary equation (22).

$$Inv_{j,l}(y_{j,k,l}) = \beta_1 * I_j + \gamma_1 * I_j * y_{j,k=be,l} + \varepsilon_1 * I_j * y_{j,k=ccs,l} + \delta * I_j * y_{j,k=flex,l} \quad (22)$$

(2) the calculation of the operation and maintenance cost.

We assume the lifetime of a coal plant is 40 years. Therefore, if a coal plant does not take any retrofit technology, the coal power plant itself will have its original operation and maintenance (OM) costs. BE technology will bring some additional OM costs for the plant. Its calculation should start from the retrofit year. The OM costs of the CCS technology will include carbon capture costs, transportation costs, and carbon storage costs. The calculation of these costs depends on the carbon emission amount of each coal plant. Thus, the function also needs to consider the actual operation hours. In addition, the carbon transportation cost should consider the distances between the coal power plant and the carbon storage site. These processes are described in supplementary equation (23).

$$OM_{j,l}(y_{j,k,l}) = \sum_{l=0}^{40} (\beta_2 + \beta_3) * I_j + y_{j,k=be,l} * \sum_{l=retrofit}^{40} (\gamma_2 + \gamma_3) * I_j + y_{j,k=ccs,l} * \sum_{l=retrofit}^{40} (\varepsilon_2 + dis_j * \varepsilon_3 + \varepsilon_4) * I_j * H_j * f_{j,l} * e_{p3} - y_{j,k=retire,l} * \sum_{l=retrofit}^{40} (\beta_2 + \beta_3) * I_j \quad (23)$$

(3) the calculation of the fuel cost.

The fuel cost can be divided into the costs before the decision-making year and after the decision-making year. Before the decision-making year, the costs will be sourced from the purchase of coal. But after the decision-making year, if the coal power plant chooses retrofit technology, the calculation of the fuel cost should consider the actual operating hours and whether to replace coal with biomass. If the power plant chooses BE technology, the purchase amount and the cost of coal will decrease, but biomass fuel will incur additional fuel costs. The cost of biomass fuel includes the purchase cost of biomass resources, pretreatment cost, field transportation cost, and the road transportation cost between the biomass collection points (BCPs) and the power plant. Therefore, the resource availability near the plant will influence whether it will choose BE technology as it will change the cost of the BE technology. If the transportation distance between the BCPs and the coal plant is short, the cost of BE will decrease greatly. Compulsory retirement will also change the fuel cost. If a coal plant chooses compulsory retirement, fuel costs will not exist after the decision-making year. These processes are described in supplementary equation (24).

$$Fuel_{j,l}(x_{i,j,l}, f_{j,l}, Dis_{i,j,l}, Dis_{j,s,l}) = \sum_{l=0}^{retrofitYr} \beta_4 * I_j * H_j + \sum_{l=retrofitYr}^{40} (I_j * H_j * f_{j,l} - \sum_i x_{i,j,l}) * \beta_4 + \sum_{l=retrofitYr}^{40} (\gamma_4 + \gamma_5 + \gamma_6) * \sum_i x_{i,j,l} + \gamma_7 * \sum_i x_{i,j,l} * dis_{i,j,l} - y_{j,k=retire,l} * \sum_{l=retrofit}^{40} \beta_4 * I_j * H_j \quad (24)$$

(4) the calculation of annual carbon emission.

The emission of coal plants will be influenced by the emission reduction technology. If a coal plant chooses compulsory retirement, it will not produce other emissions. However, if a coal plant chooses BE technology, the biomass consumption it takes can offset the equivalent emission from coal as biomass can be regarded as net-zero fuel. If a plant chooses the CCS technology, its emission can be captured with a proportion of leakage. Importantly, if a coal plant chooses BE and CCS at the same time, or chooses one before the decision-making years and then chooses another in the following decision-making year, the carbon emission absorbed by biomass during its growth can be regarded as negative emission. These processes are described in supplementary equation (25).

$$Emiss_{j,l}(y_{j,be,l}, y_{j,ccs,l}, y_{j,retire,l}, f_{j,l}, I_j, H_j, x_{i,j,l}) = f_{j,l} * I_j * H_j * e_{p3} * (1 - y_{j,retire,l}) - \sum_{l=0}^{retrofitYr} y_{j,be,l} * \sum_i x_{i,j,l} * e_p - \sum_{l=0}^{retrofitYr} y_{j,ccs,l} * (\sum_i x_{i,j,l}) * e_b - \sum_{l=0}^{retrofitYr} y_{j,ccs,l} * (f_{j,l} * I_j * H_j - \sum_i x_{i,j,l}) * e_p * e_{ccs} \quad (25)$$

Supplementary Method 3: Transition costs of the coal power sector

Many studies assessed the stranded asset of compulsory retiring coal power plants^{14–17}. The concept of the stranded asset refers to the loss of expected economic return caused by coal power plant retirement in advance. Net Present Value (NPV) is a financial metric used to evaluate the profitability of an investment and can be used as a way to assess the stranded asset. By comparing the NPV differences between the business-as-usual scenario (BAU) and the low-carbon transition scenarios, we can observe the decline in the value of coal plants' assets due to CR.

When discussing retrofit technologies, some studies use the levelized cost of electricity (LCOE) to reflect the cost increases incurred by the retrofit process, including new investment, new operation and maintenance (OM) costs, and new fuel costs. The calculation of LCOE allows for direct comparison with electricity prices directly.

By observing the calculation equation of the NPV, we find that the NPV calculation equation is also applicable for estimating the additional costs incurred by retrofit technologies. This is because the LCOE method and NPV method share the same principle of calculating future cash flows for coal plants. Therefore, to unify the economic assessment of compulsory retirement and other retrofit technologies, we extend the NPV method to reflect the transition costs of coal plants and use it to assess the cost-effectiveness of different CERPs. The calculation functions are provided in equation (26) to (30).

(1) The definition of the stranded asset.

According to the research by Wang et al.¹⁹, stranded assets represent the economic loss caused by the reduction of net cash flow during the expected life compared with the BAU scenario, and its calculation function is described in supplementary equation (26) to (28).

$$SA = NPV_{BAU} - NPV_{CR} \quad (26)$$

$$NPV_{BAU} = \sum_{l=0}^{40} \frac{R_l - C_l}{(1+r)^l} = \sum_{l=0}^{40} \frac{I_j * H_{j,l} * P - (Inv_{j,l} + OM_{j,l} + F_{j,l} - Dep_{j,l})}{(1+r)^l} \quad (27)$$

$$NPV_{CR} = \sum_{l=0}^{RetireYr} \frac{R_l - C_l}{(1+r)^l} = \frac{\sum_{l=0}^{RetireYr} I_j * H_{j,l} * P - (Inv_{j,l} + \sum_{l=0}^{RetireYr} (OM_{j,l} + Fuel_{j,l} - Depr_{j,l}))}{(1+r)^l} \quad (28)$$

Where, SA represents the stranded asset; NPV_{BAU} and NPV_{CR} represent the net cash flow under the BAU scenario and low-carbon transition scenario only with compulsory retirement; R_l represents the revenues in the year l during its operation period; C_l represents the costs of the coal plants in year l under its operation period.

(2) Expand the definition of the stranded assets and redefine it as the transition cost.

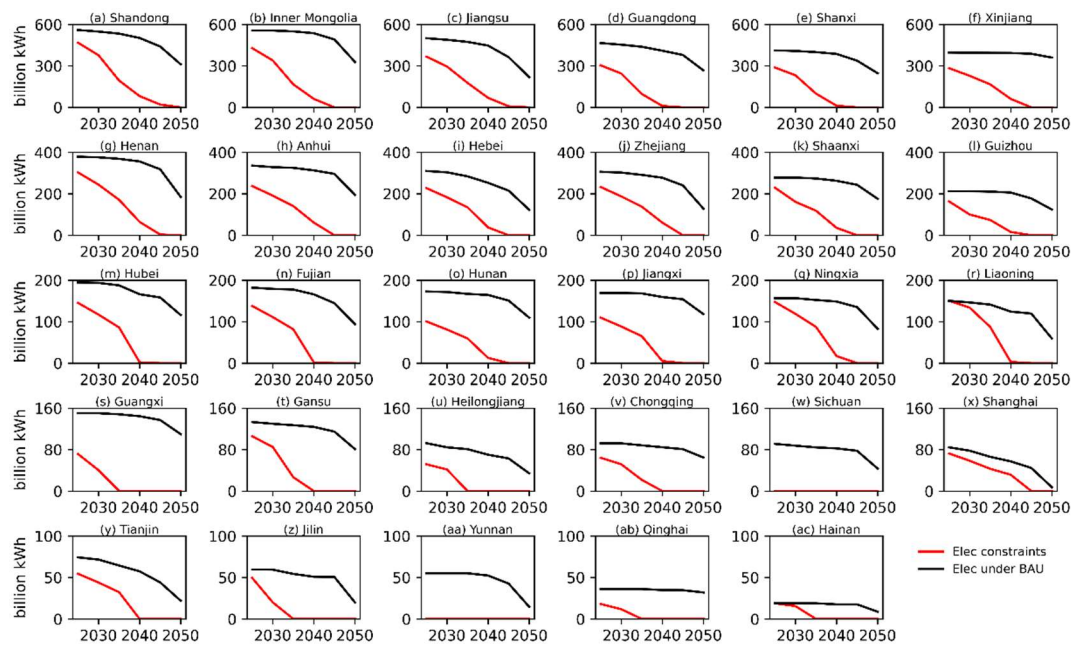
The calculation of NPV under the BAU scenarios is the same as above. While under the CERP with multiple emission reduction technologies, the calculation will be changed a little. For example, for the calculation of revenue, we need to consider the changes in operating hours brought by compulsory retirement and flexibility operation. In terms of the calculation of economic costs, we will take the changes in investment costs, operating costs, and fuel costs into account. Only in this situation, we can include BE, CCS, flexibility operation, and compulsory retirement into the same calculation framework for assessing the economic cost loss brought by the low carbon transition of the coal power plants. The calculation functions are described in supplementary equation (29) to (30).

$$TC = NPV_{BAU} - NPV_{CERP} \quad (29)$$

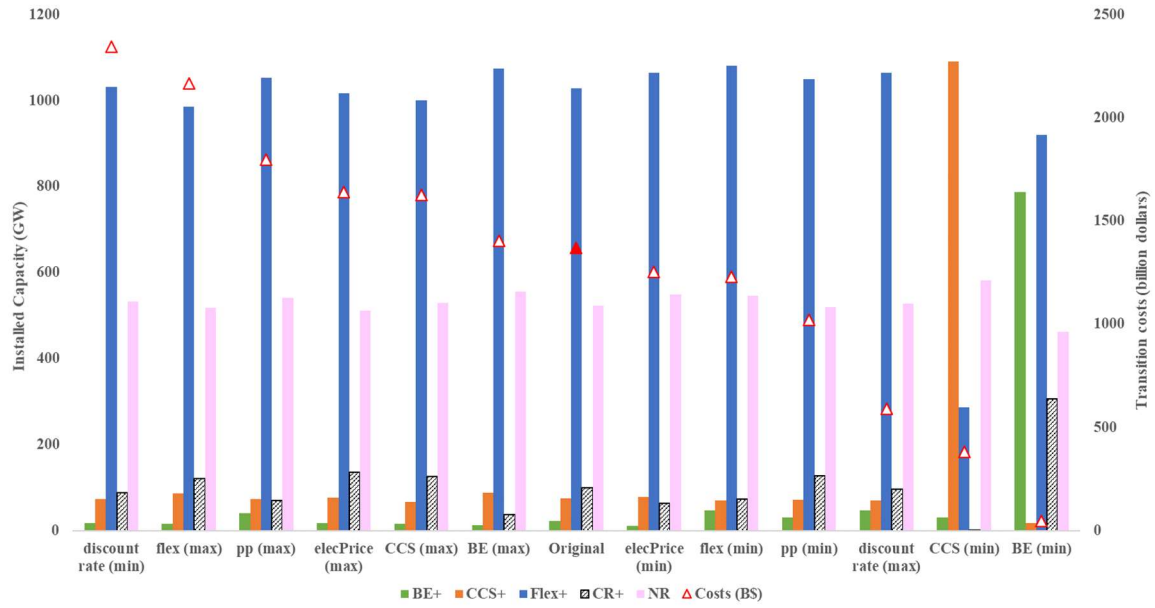
$$NPV_{CERP} = \sum_{l=0}^{40} \frac{Rev_l^{CERP} - Cost_l^{CERP}}{(1+r)^l} = \sum_{l=0}^{40} \frac{I_j * H_j * P * f_{j,l} - (Inv_{j,l}(y_{j,k,l}) + OM_{j,l}(y_{j,k,l}) + Fu_{j,l}(x_{i,j,l}, f_{j,l}, Dis_{i,j,l}, Dis_{j,s,l}, H_j, I_j) - \beta_1 * \beta_5 * I_j)}{(1+r)^l} \quad (30)$$

Where, TC represents the low-carbon transition costs of the coal power plants; NPV_{CERP} represents the net cash flow of the CERP; Other annotations can be found in the Annotation section.

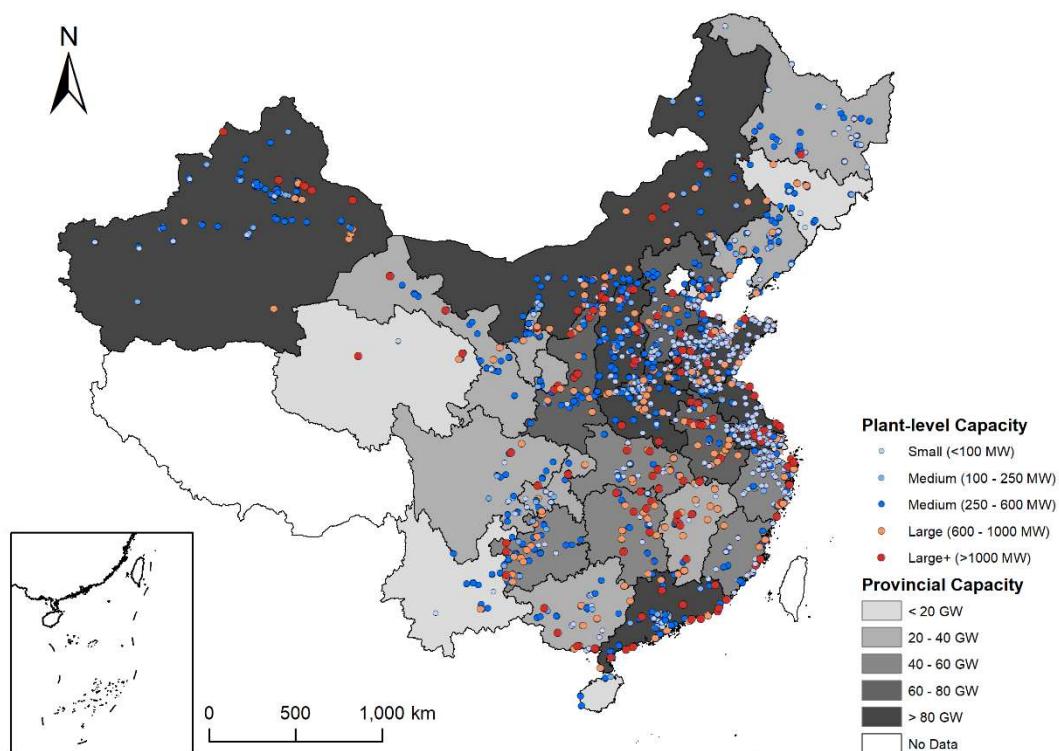
Supplementary Figures



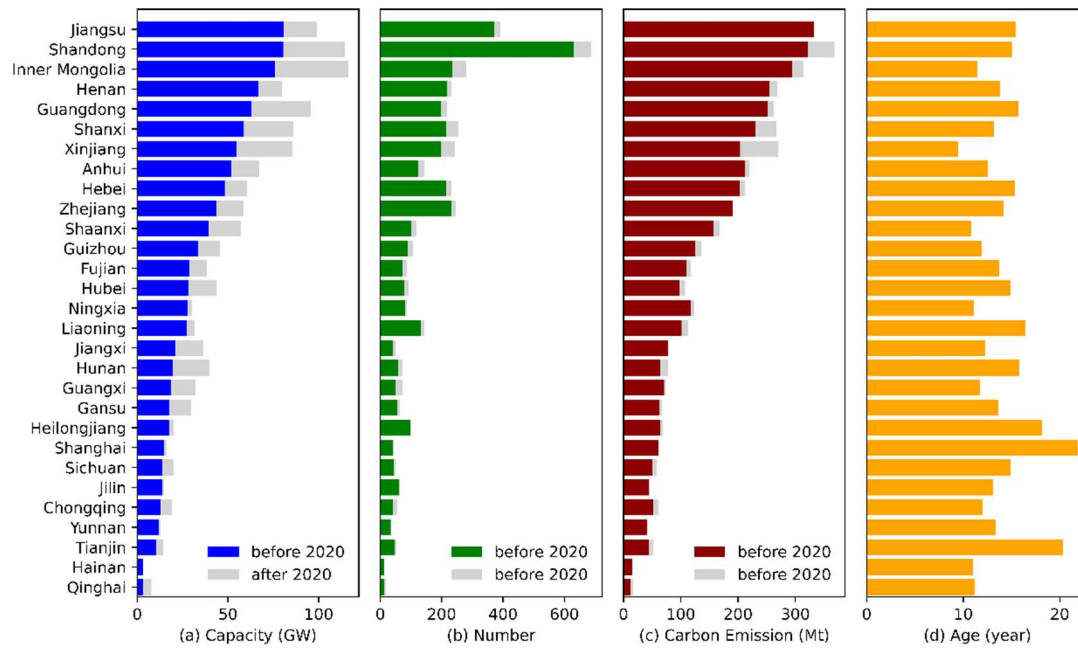
Supplementary Figure 6 Coal power generation constraints for each province. ‘Elec constraints’ in all panels represent the minimum electricity demand for coal plants, which are generated by Zhou et al³. These electricity demand considered China’s carbon neutrality target, the renewable potential, current power system configuration in China. ‘Elec under BAU’ represents the electricity generation of the coal plant if they are retired naturally when they reach their designed lifetime.



Supplementary Figure 7 Uncertainty analysis for the techno-economic parameters. Discount rate (min) and discount rate (max) represent scenarios where the model adjusts the discount rate from 5% to 3% and 8%, respectively, under the '30lp-post50zero' carbon emission reduction pathway. The bars indicate the installed capacity of coal plants opting for biomass and coal co-firing (BE), carbon capture and storage (CCS), flexibility operation (Flex), compulsory retirement (CR), and natural retirement (NR). The red triangle represents the transition costs of the coal plants, calculated based on the decision-making outcomes and corresponding parameters. PP (min), PP (max), BE (min), BE (max), CCS (min), CCS (max), flex (min), and flex (max) illustrate the plant-level decision-making results when techno-economic parameters, such as investment costs and O&M costs, are adjusted to their minimum and maximum values. ElecPrice (min) and ElecPrice (max) are used to examine the impact of electricity price on the decision-making processes. 'Original' refers to the default '30lp-post50zero' scenario discussed in the main text.



Supplementary Figure 8 Spatial distribution of coal power plants in China. The color of each point indicates the installed capacity of each coal power plant. The background color represents the total installed capacity of each province based on the spatial distribution of coal power plants.



Supplementary Figure 9 The characteristics of coal power plants in each province. The bars in the panels (a) to (c) denote the total installed capacity, the total number of plants, and the committed cumulative carbon emissions for each province before and after 2020. Panel (d) represents the average age of the coal plants in each province by the end of 2020.

Supplementary Tables

Supplementary Table 2 The statistics of coal plants for each province.

Province	Installed capacity before 2020 (GW)	Number before 2020	Installed capacity after 2020 (GW)	Number after 2020 (including 2020)
Inner Mongolia	75.71	235	40.50	45
Shandong	80.45	630	34.10	58
Guangdong	62.98	198	32.66	21
Xinjiang	54.77	199	30.80	45
Shanxi	58.80	216	27.06	38
Hunan	19.69	58	20.18	15
Jiangsu	80.63	372	18.27	19
Shaanxi	39.52	101	17.72	18
Hubei	28.22	78	15.69	15
Jiangxi	21.14	42	15.42	9
Anhui	51.92	124	15.40	19
Zhejiang	43.77	232	14.80	13
Guangxi	18.72	51	13.28	21
Henan	66.52	218	12.99	14
Hebei	48.24	215	12.13	17
Guizhou	33.62	90	12.12	17
Gansu	17.76	55	11.75	9
Fujian	28.61	73	9.70	14
Sichuan	13.95	45	6.00	5
Chongqing	13.08	42	5.96	12
Liaoning	27.10	131	4.75	13
Qinghai	3.19	12	4.62	5
Tianjin	10.38	46	3.93	6
Heilongjiang	17.48	98	2.52	3
Ningxia	27.97	81	2.10	5
Shanghai	14.62	42	1.58	3
Jilin	13.82	62	1.05	2
Yunnan	12.08	34	0.70	1
Hainan	3.34	12	0.00	0

Supplementary Table 3 Techno-economic parameters for the plant-level optimization model.

	Item		Annotation	Unit	Value (median of the range)	Range	Reference
Coal power plant	CAPEX	investment cost	β_1	\$/kw	3274	1398.5-5150.0	[²¹⁻²⁶]
	O&M	fixed O&M	β_2	\$/kw	41.66	29.3-54.0	[^{21,23,27-29}]
		variable O&M	β_3	\$/kw	31.35	24.0-38.7	[^{21,23,26-29}]
	Fuel	coal price	β_4	\$/MWh	11.45	1.9-21	[^{1,24,29-33}]
Biomass and coal co-firing (BE)	CAPEX (retrofit /added)	investment cost	γ_1	\$/kw	327.2	140.7-513.7	[^{22,25,26,29,34}]
	O&M	fixed O&M	γ_2	\$/kw	18.85	5.5-32.2	[^{27,29,35,36}]
		variable O&M	γ_3	\$/kw	27.5	25.0-30.0	[^{27,29,35,36}]
	Fuel	purchase cost	γ_4	\$/MWh	10.3	2.12-18.47	[^{1,24,25,27-30,34-39}]
		pretreatment cost	γ_5	\$/MWh	6.15	2.4-9.9	[^{1,30,34,37,38,40}]
		transportation cost(ax+b): a	γ_7	\$(/MWh*km)	0.065	0.00045-0.13	[^{1,27,30,34,36,38-40}]
		transportation cost(ax+b): b	γ_6	\$/MWh	6.75	0.33-13.17	[^{1,27,30,34,36,38-40}]
Carbon capture and storage (CCS)	CAPEX (retrofit /added)	investment cost	ε_1	\$/kw	1559.75	205-2914.5	[^{22,24,25,41,42}]
	Carbon capture	equipment capture cost	ε_2	\$/t CO2	40	29-51	[⁴³⁻⁴⁶]
	Carbon transportation	transportation cost	ε_3	\$(/t CO2*km)	0.1195	0.015-0.224	[^{23,41,43-48}]
	Carbon storage	storage cost	ε_4	\$/t CO2	17.39	2.93-31.85	[^{23,25,30,41,43-45,48}]
Flexibility operation (Flex)	CAPEX (retrofit /added)	investment cost	δ	\$/kw	327.4	163.7-982.2	assumption
Others	Lifetime		n	year	40	25-40	
	Depreciation rate		β_5	%	5	3-8	
	Electricity price		P	\$/kWh	0.06	0.048-0.072	Policy documents and News

Supplementary Table 4 Statistics of the uncertainty analysis for the techno-economic parameters.

scenario	BE+	CCS+	Flex+	Retire	NR	Costs (Billion US\$)
discount rate(min)	17.8	72.5	1032.1	88.5	532.7	2342.5
flex (max)	15.8	86.4	986	121	517.2	2168
pp (max)	40.3	73.4	1053.5	70.4	540.8	1796.7
elecPrice (max)	17.7	76	1017	135.1	511.1	1641.3
CCS (max)	15.8	67.1	1000.6	124.9	529.2	1627.6
elecPrice (min)	10.3	78.1	1064.9	63.2	548.4	1252.1
pp (min)	29.8	72.2	1049.6	127.8	519.4	1021.4
CCS (min)	30.4	1091.4	286.2	0.7	581.2	381.4
BE (min)	786.9	18.1	920.7	306.7	461.8	47.1
discount rate(min)	17.8	72.5	1032.1	88.5	532.7	2342.5
flex (max)	15.8	86.4	986	121	517.2	2168
pp (max)	40.3	73.4	1053.5	70.4	540.8	1796.7
elecPrice (max)	17.7	76	1017	135.1	511.1	1641.3

Note: Discount rate (min) and discount rate (max) represent scenarios where the model adjusts the discount rate from 5% to 3% and 8%, respectively, under the '30lp-post50zero' carbon emission reduction pathway. PP (min), PP (max), BE (min), BE (max), CCS (min), CCS (max), flex (min), and flex (max) illustrate the plant-level decision-making results when techno-economic parameters, such as investment costs and O&M costs, are adjusted to their minimum and maximum values. ElecPrice (min) and ElecPrice (max) are used to examine the impact of electricity price on the decision-making processes. The values under the 'BE+' (biomass and coal co-firing technology), 'CCS+' (carbon capture and storage), 'Flex+' (flexibility operation), 'Retire' (compulsory retirement), and 'NR' (natural retirement) columns represent the total installed capacity of coal plants that have adopted the respective strategies over their lifespans. The final columns indicate the transition costs associated with each scenario

Supplementary Table 5 Carbon emission parameters of the coal power plants and retrofit technologies.

Process	Value	Unit	Annotation	Reference
Coal-fired power	801.7	g/kWh	e_p	[⁴⁹]
Absorption rate ^{a,b} (straw production/ carbon absorption)	375.72	g/kWh	e_b	[^{50,51}]
Carbon capture rate	90	%	e_{ccs}	[⁴⁰]

Supplementary Table 6 Coal power generation constraints for each province.

Province	2025	2030	2035	2040	2045	2050
Tianjin	54560.69	43952.48	32334.12	6.39	2.38	1.08
Hebei	227418.08	183201.33	134774.04	37718.09	3.31	1.17
Shanxi	288258.18	232212.24	99309.46	11765.47	2.98	1.15
Inner Mongolia	428603.22	339433.62	166476.14	60915.51	9.30	1.10
Liaoning	150399.27	134040.75	87350.88	3014.34	2.75	1.13
Jilin	49506.83	20026.75	92.70	2.26	2.19	1.09
Heilongjiang	51663.22	41617.96	40.49	1.87	1.77	1.06
Shanghai	72565.18	58456.38	43004.07	31469.00	3.62	1.17
Jiangsu	367100.59	295725.40	177167.89	68739.40	7832.28	1.22
Zhejiang	232881.07	187602.15	138011.57	60907.98	3.49	1.20
Anhui	237356.90	191207.75	140664.07	60909.74	4.14	1.18
Fujian	137985.99	111157.46	81773.02	1721.86	2.88	1.18
Jiangxi	109882.69	88518.27	65119.00	5154.50	2.77	1.15
Shandong	468152.50	377129.70	194207.69	81208.59	20300.51	1.21
Henan	303157.63	244214.86	171105.66	64303.25	3396.49	1.21
Hubei	145451.11	117171.11	86198.02	2170.70	2.80	1.10
Hunan	100663.78	81091.61	59655.42	12744.72	2.66	1.10
Guangdong	304176.80	245035.82	97031.97	10098.45	2.53	1.17
Guangxi	71552.40	40290.94	433.26	2.22	2.20	1.11
Hainan	19252.41	15509.13	11.36	0.78	0.77	0.72
Chongqing	64283.20	51784.31	21674.67	15.88	2.67	1.09
Sichuan	7.50	6.03	4.36	2.13	1.97	1.00
Guizhou	163186.50	100169.23	73688.57	15822.34	3.17	1.16
Yunnan	12.78	10.27	7.55	5.52	5.60	2.36
Shaanxi	229895.56	161142.00	118545.26	35684.71	2.78	1.14
Gansu	105695.05	85143.53	26759.31	5.88	2.59	1.10
Qinghai	18124.50	11763.78	87.65	4.74	2.55	1.12
Ningxia	147673.10	118961.11	87515.03	17236.33	3.46	1.19
Xinjiang	283380.76	228283.22	166470.19	60911.17	4.76	1.18

Note: the unit is GWh. This electricity demand of coal power plants is simulated by the Generation and Transmission Expansion Planning (GTEP) model developed by Zhou et al³, which considered China's carbon neutrality target, the renewable potential allocation and development, current power system configuration in China.

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